

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**CO3\_AT1 — EXPERIMENT**

|                         |                                                                                                                                                                                                  |                      |                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------|
| <b>Course Code:</b>     | DSA06                                                                                                                                                                                            | <b>Course Title:</b> | Data Handling and Visualization |
| <b>CO Assessed:</b>     | CO3 – Construction (BL3)                                                                                                                                                                         | <b>Assessment:</b>   | Assessment Tool 1 – Experiment  |
| <b>Weightage:</b>       | 20% of CIA                                                                                                                                                                                       | <b>Total Marks:</b>  | 25 (5 Questions × 5 Marks)      |
| <b>CO3 Statement:</b>   | <i>Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data.(BL3)</i> |                      |                                 |
| <b>Student Name:</b>    | PRIYANKA G                                                                                                                                                                                       | <b>Reg. No.:</b>     | 192524124                       |
| <b>Section / Batch:</b> |                                                                                                                                                                                                  | <b>Date:</b>         | 11/08/2026                      |

**Code of Conduct**

*I Priyanka.G/192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_

**Instructions**

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

**Annexure A – EXPERIMENT**

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|                          |                           |                           |
|--------------------------|---------------------------|---------------------------|
| <b>Faculty In-charge</b> | <b>Course Coordinator</b> | <b>Head of Department</b> |
| Signature: _____         | Signature: _____          | Signature: _____          |
| Date: _____              | Date: _____               | Date: _____               |

### Problem 5 — Pie Chart — Revenue Share by Product Line (5 Marks)

#### Given Data:

| Product Line | Electronics | Clothing | Books  |
|--------------|-------------|----------|--------|
| Revenue (₹)  | 50,000      | 30,000   | 20,000 |

**Task:** Construct a pie chart showing each product line's percentage share of total revenue.

#### Construction Plan (Key Elements):

|                          |                                                                              |
|--------------------------|------------------------------------------------------------------------------|
| <b>Chart Type</b>        | Pie chart (single-series, part-to-whole)                                     |
| <b>Coordinate System</b> | Polar — each slice defined by radius (constant) and angle ( $\propto$ value) |
| <b>Legend</b>            | Product Line (Electronics, Clothing, Books), one colour per slice            |
| <b>Labels</b>            | Percentage share displayed inside each slice                                 |

#### Worked Calculations / Construction Steps:

A pie chart is the correct choice for this dataset because the three product lines are nominal categories that together make up 100% of a single whole (total company revenue), and the task specifically asks for each category's percentage share of that whole — precisely the part-to-whole relationship a pie chart is designed to show. With only three categories, none of them a very thin sliver, a pie chart also avoids the readability problems (thin, hard-to-compare slices) that appear once category counts grow past five or six.

Total revenue = ₹50,000 + ₹30,000 + ₹20,000 = ₹1,00,000.

Each product line's percentage share of the total is calculated as (Product Revenue ÷ Total Revenue) × 100:

- Electronics:  $(50,000 \div 1,00,000) \times 100 = 50\%$
- Clothing:  $(30,000 \div 1,00,000) \times 100 = 30\%$
- Books:  $(20,000 \div 1,00,000) \times 100 = 20\%$

Since a full pie chart corresponds to  $360^\circ$ , each slice's angle is calculated as (Percentage ÷ 100) ×  $360^\circ$ :

- Electronics:  $0.50 \times 360^\circ = 180^\circ$
- Clothing:  $0.30 \times 360^\circ = 108^\circ$
- Books:  $0.20 \times 360^\circ = 72^\circ$

Check:  $180^\circ + 108^\circ + 72^\circ = 360^\circ$ , confirming the three slice angles correctly account for the full circle.

The chart is constructed with Electronics as the largest slice (exactly half the circle), followed by Clothing and Books, each slice labelled with its percentage and matched to its product line via the legend, along with a descriptive title and a one-line interpretation identifying Electronics as the major revenue-generating product line.

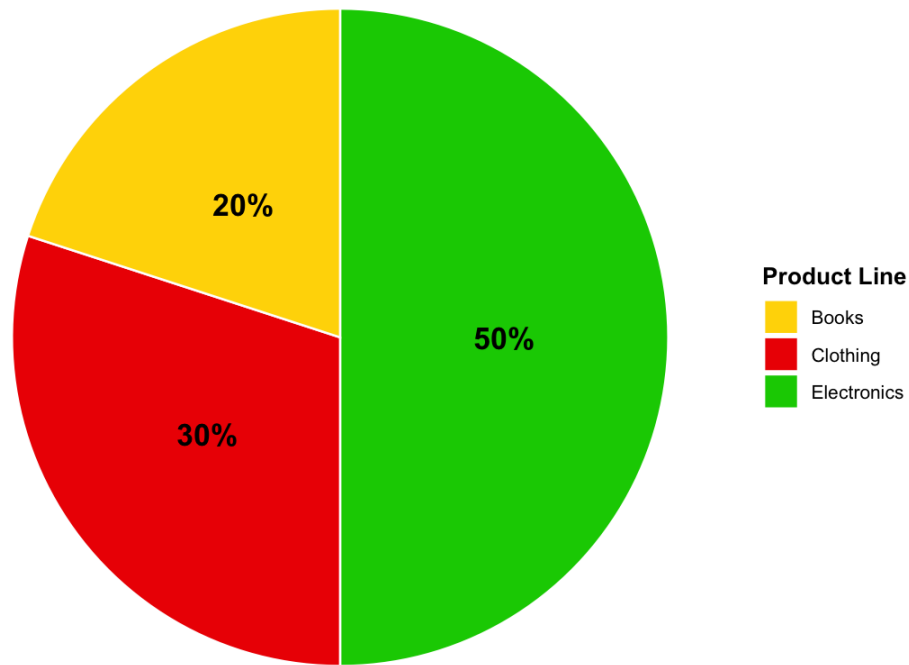
A bar chart could also display these three revenues figures and would in fact be more accurate for judging the exact difference between Clothing (30%) and Books (20%), since bar length is easier to compare precisely than slice angle. However, because the task

specifically asks for each product line's percentage share of the total — a part-to-whole relationship — a pie chart communicates the 'these three parts sum to one whole' framing more directly and immediately than a bar chart, which shows three independent magnitudes without visually reinforcing that they add up to 100%.

### Constructed Chart:

#### **192524124 Problem 5 - Revenue Share by Product Line**

Electronics contributes the highest revenue share (50%), followed by Clothing (30%) and Books (20%).



*Interpretation: Electronics is the major revenue-generating product line.*

*Figure 1: Pie Chart — Revenue Share by Product Line*

## Problem 6 — Bean Plot — Quiz Marks (5 Marks)

### Given Data:

| Student | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
|---------|----|----|----|----|----|----|----|----|----|
| Marks   | 40 | 50 | 55 | 60 | 60 | 60 | 65 | 70 | 80 |

**Task:** Construct a bean plot for this data and identify any repeated values that become visible in the plot.

### Construction Plan (Key Elements):

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Chart Type  | Bean plot (individual data points + density/violin outline in one axis) |
| Y-Axis      | Quiz Marks — numeric, quantitative                                      |
| Key Feature | Overlapping/repeated points reveal the distribution's density directly  |
| Annotation  | Callout marking the repeated value (60, occurring 3 times)              |

### Worked Calculations / Construction Steps:

A bean plot is the appropriate chart for this dataset because it is specifically designed to reveal individual data points and any repeated or clustered values within a small sample, which a standard box plot (which only shows five summary numbers) or a bar chart (which is not suited to a single numeric distribution) would hide. With only 9 observations, plotting every raw value directly, rather than summarising them away, is both feasible and the most informative choice.

Sorted data: 40, 50, 55, 60, 60, 60, 65, 70, 80 ( $n = 9$ ).

Mean =  $(40+50+55+60+60+60+65+70+80) \div 9 = 540 \div 9 = 60$ .

Median = the 5th value in the sorted list of 9 = 60.

Mode = 60 (the only value that repeats, occurring 3 times, out of 9 total observations).

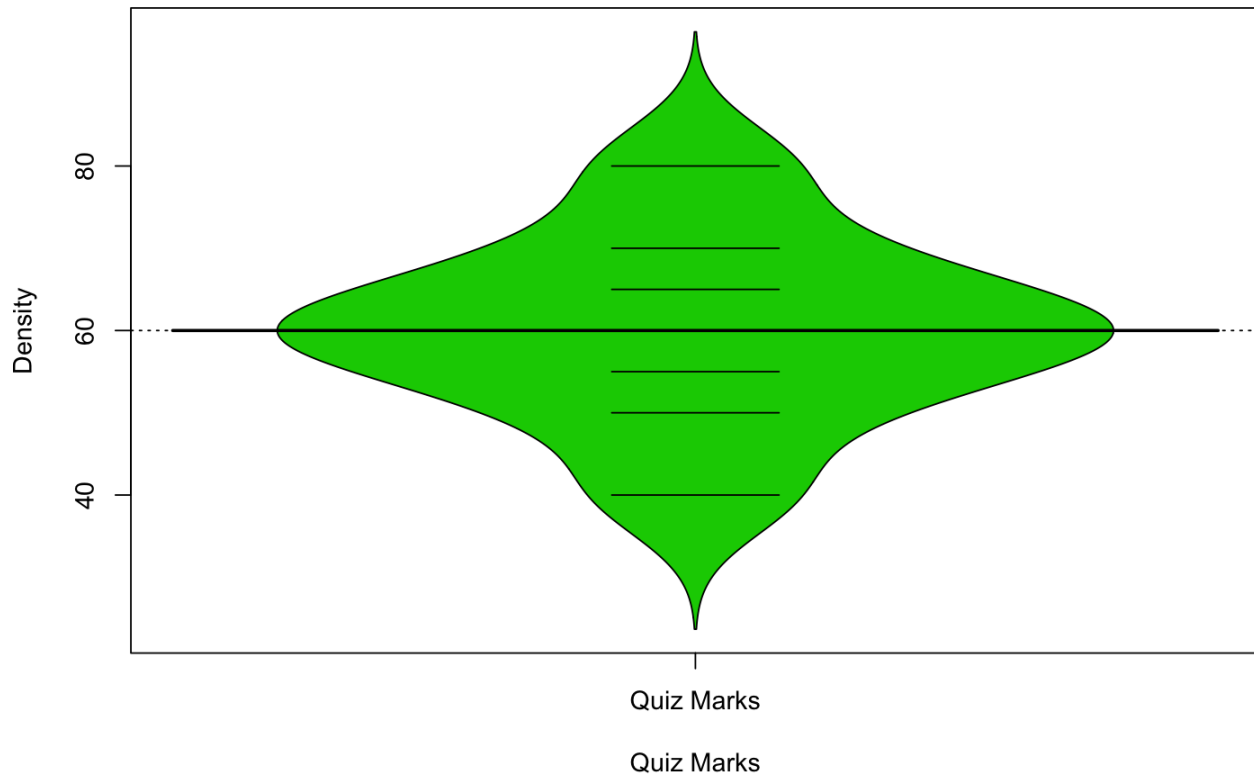
A bean plot is constructed by plotting every individual observation as a point along the marks axis (rather than summarising them into a box), so that repeated values naturally stack or cluster at the same height, directly showing how often each value occurs; a smoothed density/violin outline is drawn around the points to show the overall shape of the distribution.

Because the mean, median, and mode all equal exactly 60 in this dataset, the distribution is centred and fairly symmetric, and the bean plot's density curve is widest at 60, directly reflecting the repeated cluster of three students who all scored 60 — the plot is explicitly annotated to call this out ('Repeated value: 60 occurs 3 times') so the repetition is not left for the viewer to infer only from the shape.

A box plot would still correctly report the median (60) and IQR for this data, but it discards individual observations in favour of five summary numbers, so the fact that three separate students tied at exactly 60 marks would be completely invisible on a box plot — that repetition is only preserved because the bean plot retains every raw data point.

### Constructed Chart:

### 192524124 Problem 6 - Bean Plot of Quiz Marks



*Interpretation: Nine quiz observations are shown. The repeated score of 60 occurs three times.*

*Figure 2: Bean Plot — Quiz Marks, with Repeated Value (60) Annotated*

## Problem 12 — Violin Plot — Class Marks Comparison (5 Marks)

### Given Data:

| Class A<br>(Clustered) | 50 | 55 | 58 | 60 | 60 | 62 | 65 |
|------------------------|----|----|----|----|----|----|----|
| Class B<br>(Spread)    | 30 | 40 | 50 | 60 | 70 | 80 | 90 |

**Task:** Construct violin plots for both classes and compare their shapes, even though both classes have a similar average mark.

### Construction Plan (Key Elements):

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| Chart Type  | Violin plot (density shape) with an inner box plot and raw points overlaid         |
| X-Axis      | Class — nominal category (Class A, Class B)                                        |
| Y-Axis      | Marks — numeric, quantitative                                                      |
| Key Insight | Same central tendency, very different spread — visible only via the violin's width |

### Worked Calculations / Construction Steps:

A violin plot is the correct choice here specifically because the task requires comparing the shape and spread of two distributions, not just their central tendency. A simple bar chart of the two class averages would show two nearly identical bars (since both medians equal 60) and would completely hide the fact that the two classes are distributed very differently — exactly the kind of masking a violin plot's density-shape encoding is designed to prevent.

Class A (sorted): 50, 55, 58, 60, 60, 62, 65 ( $n = 7$ ). Median = 4th value = 60. Lower half (50, 55, 58) →  $Q1 = 55$ . Upper half (60, 62, 65) →  $Q3 = 62$ .  $IQR = 62 - 55 = 7$ . Range =  $65 - 50 = 15$ .

Class B (sorted): 30, 40, 50, 60, 70, 80, 90 ( $n = 7$ ). Median = 4th value = 60. Lower half (30, 40, 50) →  $Q1 = 40$ . Upper half (70, 80, 90) →  $Q3 = 80$ .  $IQR = 80 - 40 = 40$ . Range =  $90 - 30 = 60$ .

Both classes share an identical median of 60 — if only a single summary statistic were reported, the two classes would look the same. However, Class A's IQR (7) and range (15) are far smaller than Class B's IQR (40) and range (60), meaning Class A's marks are tightly clustered around the median while Class B's marks are spread across a much wider band.

The violin plot makes this difference immediately visible without needing to read the numbers: Class A is drawn as a narrow, tall shape concentrated near 55–65, while Class B is drawn as a much wider, flatter shape stretching from roughly 30 to 90 — demonstrating exactly why a violin plot is preferred over reporting the mean/median alone when the question specifically concerns comparing distribution shape and spread, not just central tendency.

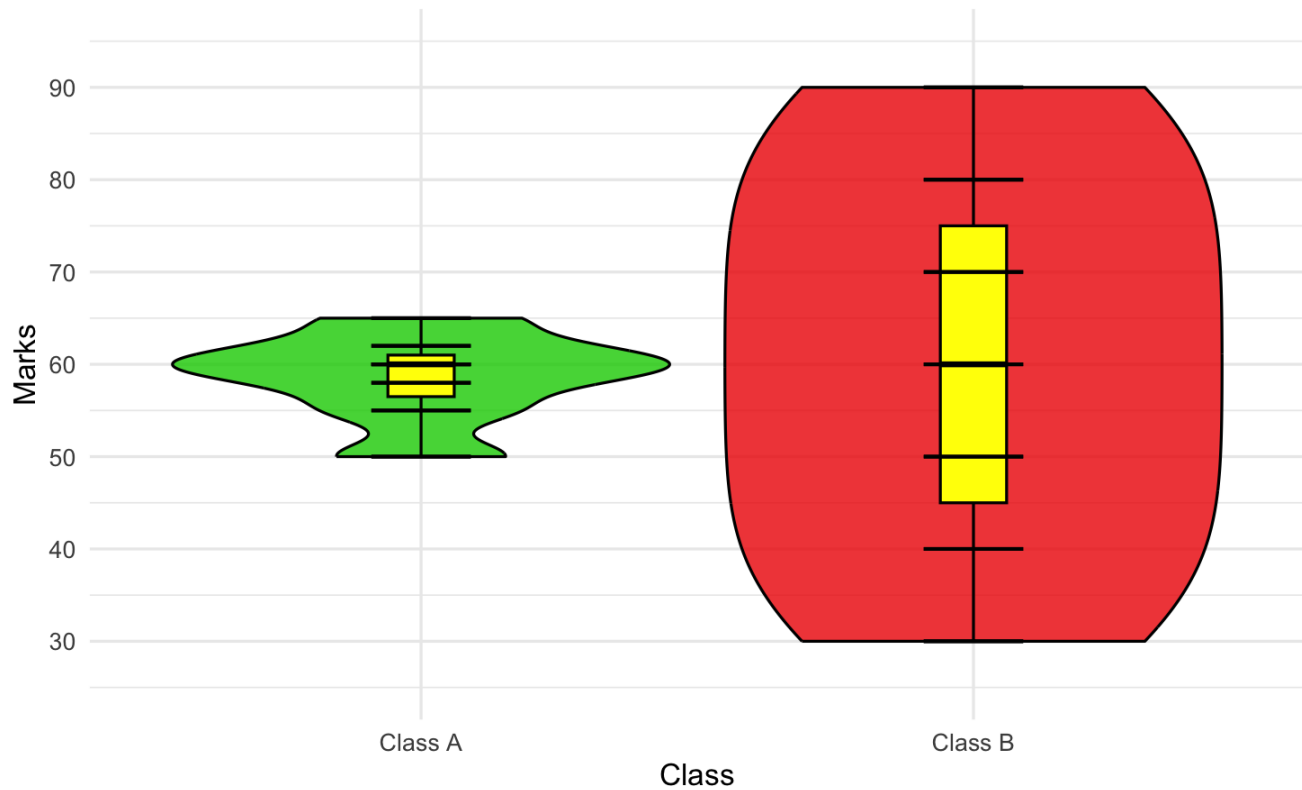
A plain box plot would also correctly show Class B's larger IQR, but a box plot only encodes four boundary points ( $Q1$ , median,  $Q3$ , and the whiskers) and cannot show whether the underlying data is smoothly spread across that range or clustered into sub-groups within it. The violin plot's continuous density curve carries strictly more shape information than the

box plot alone, which is why the inner box is kept as a secondary reference layer inside each violin rather than used as the primary chart.

### Constructed Chart:

#### 192524124 Problem 12 - Violin Plot of Class Marks

Class A is clustered, whereas Class B is widely spread



*Interpretation: Both classes have similar average marks, but Class B has greater variability.*

*Figure 3: Violin Plot — Class A (Clustered) vs. Class B (Spread)*

## Problem 18 — Box Plot — Test Scores Comparison (5 Marks)

### Given Data:

| Section A | 60 | 65 | 68 | 70 | 72 | 75 | 80 |
|-----------|----|----|----|----|----|----|----|
| Section B | 40 | 50 | 55 | 60 | 65 | 70 | 95 |

**Task:** Construct box plots for both sections and compare their medians and spread.

### Construction Plan (Key Elements):

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| <b>Chart Type</b>  | Box plot (five-number summary), one box per section                             |
| <b>X-Axis</b>      | Section — nominal category (Section A, Section B)                               |
| <b>Y-Axis</b>      | Test Score — numeric, quantitative                                              |
| <b>Key Feature</b> | Median line, IQR box, whiskers, and any outliers beyond $1.5 \times \text{IQR}$ |

### Worked Calculations / Construction Steps:

A box plot is the appropriate chart for this task because it is built directly from the five-number summary (minimum, Q1, median, Q3, maximum) and is the standard tool for comparing the median and spread of two or more groups side by side, which is exactly what the task asks for — comparing medians and spread between Section A and Section B — without needing to plot every individual point as a violin or bean plot would.

Section A (sorted): 60, 65, 68, 70, 72, 75, 80 ( $n = 7$ ). Median = 4th value = 70. Lower half (60, 65, 68) → Q1 = 65. Upper half (72, 75, 80) → Q3 = 75. IQR =  $75 - 65 = 10$ . Whisker fences:  $Q1 - 1.5 \times \text{IQR} = 65 - 15 = 50$ ;  $Q3 + 1.5 \times \text{IQR} = 75 + 15 = 90$ . Since min (60) and max (80) both fall inside  $[50, 90]$ , Section A has no outliers.

Section B (sorted): 40, 50, 55, 60, 65, 70, 95 ( $n = 7$ ). Median = 4th value = 60. Lower half (40, 50, 55) → Q1 = 50. Upper half (65, 70, 95) → Q3 = 70. IQR =  $70 - 50 = 20$ . Whisker fences:  $Q1 - 1.5 \times \text{IQR} = 50 - 30 = 20$ ;  $Q3 + 1.5 \times \text{IQR} = 70 + 30 = 100$ . The maximum value (95) falls just inside the upper fence (100), so by the strict  $1.5 \times \text{IQR}$  rule it is not a formal statistical outlier, even though it sits visually well above the rest of Section B's scores.

Comparing the two boxes: Section A's median (70) is noticeably higher than Section B's median (60), and Section A's IQR (10) is exactly half of Section B's IQR (20) — meaning Section A both scored higher on average and was more tightly clustered, while Section B's wider box and taller whisker reflect a more spread-out, less consistent set of scores.

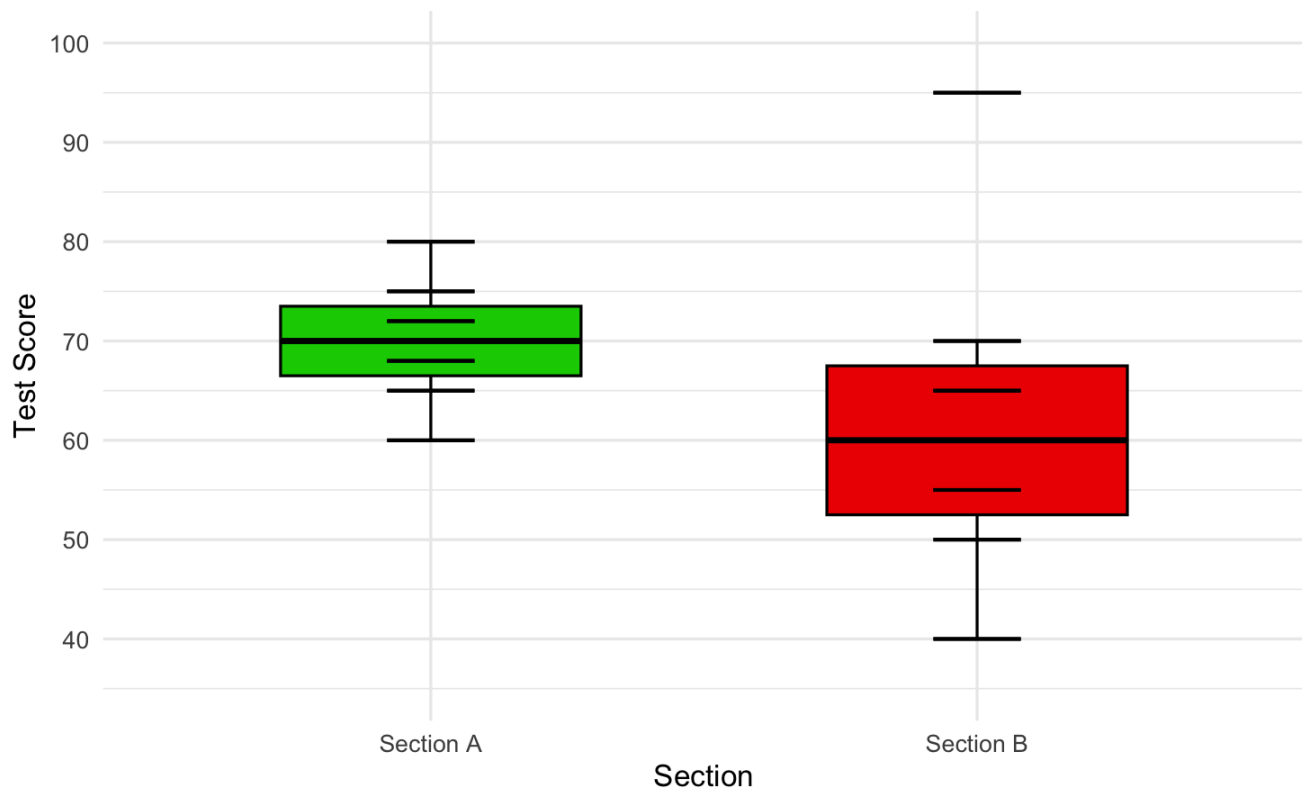
A violin plot could also be used for this comparison and would add information about the density shape within each section, but since the task specifically asks to compare medians and spread — quantities defined directly by the five-number summary — a box plot is the more direct, standard tool for this particular comparison, and its explicit quartile boundaries make reading off Q1, median, and Q3 for each section unambiguous in a way a smoothed density curve does not.

### Constructed Chart:



## 192524124 Problem 18 - Box Plot of Test Scores

Comparison of median and spread between Section A and Section B



*Interpretation: Section A has a median of 70, while Section B has a median of 60 and greater overall spread.*

*Figure 4: Box Plot — Section A vs. Section B Test Scores*

## Problem 24 — Stacked Bar Chart — Pass/Fail Composition (5 Marks)

### Given Data:

| Class   | Pass | Fail | Total |
|---------|------|------|-------|
| Class A | 35   | 5    | 40    |
| Class B | 28   | 12   | 40    |

**Task:** Construct a stacked bar chart comparing total students and the pass/fail composition of both classes.

### Construction Plan (Key Elements):

|            |                                                                |
|------------|----------------------------------------------------------------|
| Chart Type | Stacked bar chart (two segments per bar)                       |
| X-Axis     | Class — categorical (Class A, Class B)                         |
| Y-Axis     | Number of Students — numeric, cumulative                       |
| Legend     | Result — Pass (green, bottom segment), Fail (red, top segment) |

### Worked Calculations / Construction Steps:

A stacked bar chart is the correct choice because the task asks for two things simultaneously: the total number of students in each class, and how that total breaks down into Pass and Fail — a stacked bar naturally encodes both, since the full bar height shows the total while the segment boundary within it shows the sub-category split, whereas a simple (non-stacked) bar chart of pass counts alone would lose the fail information, and a pie chart would lose the ability to compare totals across the two classes.

Class A total = Pass (35) + Fail (5) = 40 students. Pass rate =  $(35 \div 40) \times 100 = 87.5\%$ . Fail rate =  $(5 \div 40) \times 100 = 12.5\%$ .

Class B total = Pass (28) + Fail (12) = 40 students. Pass rate =  $(28 \div 40) \times 100 = 70\%$ . Fail rate =  $(12 \div 40) \times 100 = 30\%$ .

Both classes have the same total enrolment (40 students each), which is itself a useful check: it means the two bars in the stacked chart are drawn to the same total height, so any visual difference between them comes entirely from how that height is divided between the Pass and Fail segments — not from a difference in class size.

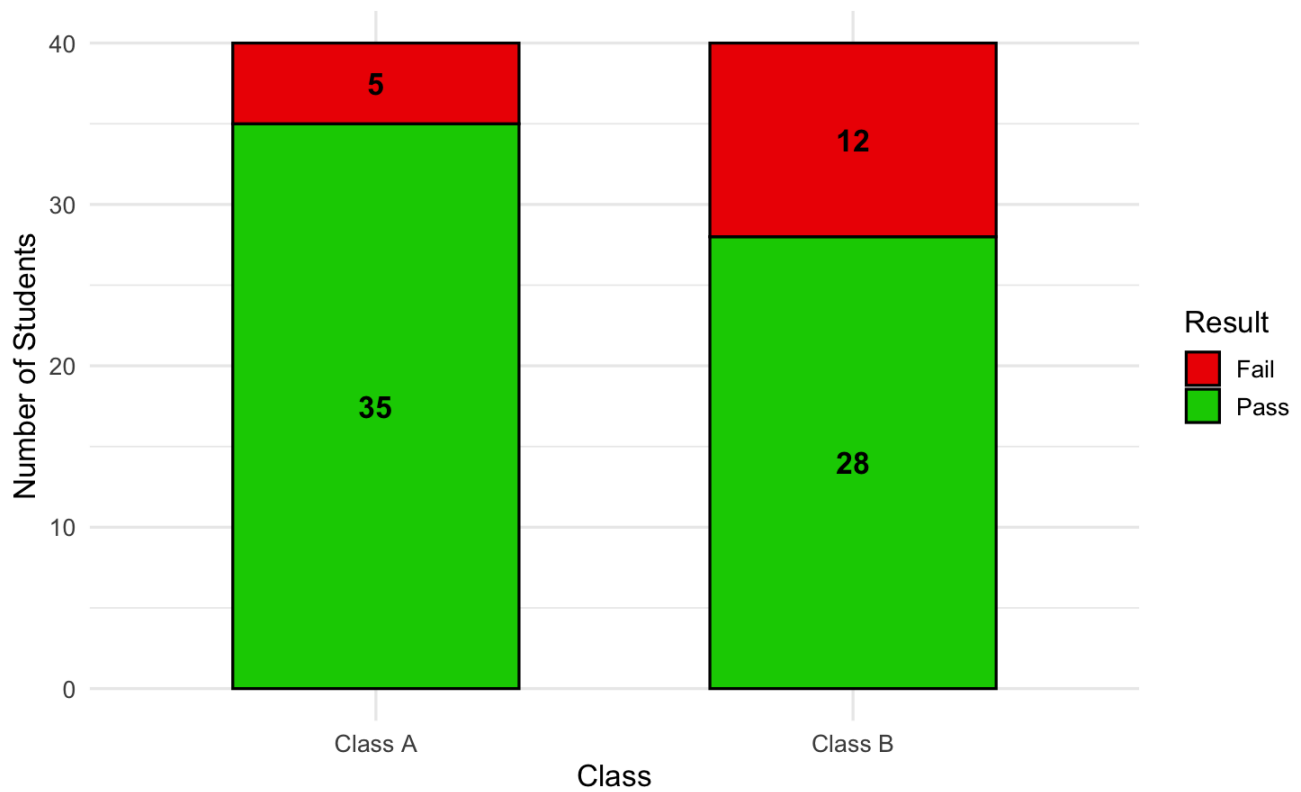
Each bar is constructed bottom-up: the Pass segment is drawn first, from 0 up to the pass count (0–35 for Class A, 0–28 for Class B), and the Fail segment is stacked directly on top of it, from the pass count up to the total (35–40 for Class A, 28–40 for Class B). Comparing segment heights directly shows that Class A's pass rate (87.5%) is substantially higher than Class B's (70%), and correspondingly Class B's fail segment is more than twice the size of Class A's in absolute terms (12 vs. 5 students).

A grouped (side-by-side) bar chart, with separate adjacent Pass and Fail bars for each class, was also considered, and would make the exact Fail counts (5 vs. 12) marginally easier to compare directly. However, it would not show each class's total enrolment as a single bar height the way the stacked design does, and since both classes happen to share the same total (40 students), the stacked chart's equal-height bars make that equality immediately visible — a fact a grouped bar chart would obscure.

### Constructed Chart:

## 192524124 Problem 24 - Pass and Fail Composition by Class

Class A has a higher pass proportion than Class B.



Interpretation: Class A has 35 passes and 5 failures, while Class B has 28 passes and 12 failures.

Figure 5: Stacked Bar Chart — Pass and Fail Composition by Class